DOCTORAL THESIS

Solution Combustion Synthesis of Copper Nanorods/Nanowires and Their Surface Modification for Enhanced PTFE Composites

Abstract

Polytetrafluoroethylene (PTFE) is widely used as a sealing and bearing material due to its exceptional chemical inertness and low friction coefficient, but its inherent cold flow tendency and extremely low thermal conductivity severely limit its application in load-bearing and heat-dissipation scenarios. This thesis begins with the preparation of copper nanorods/nanowires via solution combustion synthesis (SCS), systematically investigating combustion synthesis systems using copper nitrate as the oxidizer and hexamethylenetetramine (HMT), citric acid, or sucrose as reducing agents, and revealing the regulatory mechanisms of fuel type and fuel-to-oxidizer ratio on product phase and morphology. To address the core issue of weak interfacial bonding between copper nanofillers and the PTFE matrix, stearic acid was employed for surface modification of copper nanostructures, constructing a composite system with robust interfacial bonding. PTFE was surface-activated via plasma treatment, and changes in surface wettability and chemical composition were systematically characterized. On this basis, copper nanorod/nanowire-filled PTFE composites were prepared, and their compressive properties, tribological performance, and thermal conductivity were systematically tested. Results demonstrate that the thermal conductivity of stearic acid-modified copper nanowire-filled PTFE composites reaches 0.41 W/m·K (10 vol% filling), representing an approximately 105% improvement over pure PTFE; compressive strength is enhanced by 30–70%. This thesis provides systematic theoretical foundations and experimental data for the design of high-performance PTFE-based thermally conductive and wear-resistant composites.

Keywords: Solution combustion synthesis; Copper nanowires; Stearic acid modification; PTFE composites; Thermal conductivity; Compressive properties

Chapter 1 Introduction

1.1 Research Background

PTFE is known as the "king of plastics," with C–F bond energy as high as 485 kJ/mol, conferring exceptional chemical stability and thermal stability. However, the PTFE molecular chain adopts a helical conformation, with the carbon backbone tightly enveloped by a shell of fluorine atoms, resulting in extremely low surface energy (approximately 18 mN/m) and making interfacial bonding with other materials extremely difficult. In tribological applications, the wear rate of pure PTFE reaches the order of 10^{-3} $mm^3/N \cdot m$, and its cold flow tendency causes irreversible deformation under sustained loads.

Copper nanostructures, particularly one-dimensional copper nanowires (CuNWs), are considered ideal polymer thermal conductive fillers due to their high aspect ratio, excellent electrical and thermal conductivity, and relatively low cost. Theoretical studies indicate that one-dimensional nanowires outperform zero-dimensional nanoparticles in constructing continuous thermal conductive networks, as nanowires more readily form bridging structures, thereby establishing low-thermal-resistance heat transfer pathways.

However, applying copper nanowires to PTFE matrices faces dual challenges: first, copper nanowires are susceptible to oxidation in air, and high-temperature processing (PTFE sintering requires above 370°C) leads to severe oxidation; second, the surface energy mismatch between copper and PTFE results in weak interfacial bonding, causing interfacial thermal resistance that severely limits the enhancement of thermal conductivity.

1.2 Research Progress on Solution Combustion Synthesis

Solution combustion synthesis (SCS) is a rapid, energy-efficient nanomaterial preparation technique based on exothermic redox reactions between oxidizers (typically metal nitrates) and fuels (such as glycine, urea, citric acid) . SCS starts from aqueous solutions, ensuring molecular-level mixing of reactants, and the driving force for lattice formation derives from internal chemical energy . The temperature input required for SCS is not high—it is only needed to trigger the combustion reaction, rather than providing continuous energy input as in conventional material preparation routes .

The key parameters of SCS include fuel type, fuel-to-oxidizer ratio (φ), and pH value of the solution . The propellant chemistry theory is commonly used to calculate the stoichiometry of fuel and oxidizer . When φ = 1, it represents the stoichiometric state, where atmospheric oxygen is not required for complete oxidation of the fuel; φ > 1 and φ < 1 indicate fuel-rich and fuel-lean conditions, respectively .

Research has shown that oxygen-free fuels containing amine groups (such as hydrazine or hexamethylenetetramine) tend to form highly reactive intermediate complexes with oxidizers . The solution combustion with these fuels may exhibit a distinctive mechanism compared to other fuels. For the $Ni(NO_3)_2$ + hexamethylenetetramine system, fuel-rich gels (φ = 1.5) exhibit significantly higher heating rates (approximately 550 K/s) and Tmax (1475 K) compared to fuel-lean gels (φ = 0.75, heating rate approximately 100 K/s, Tmax approximately 970 K) .

Thermodynamic calculations based on Gibbs free energy minimization principles reveal that for the $Ni(NO_3)_2$-glycine/hexamethylenetetramine systems, T_ad increases with increasing φ value, reaching a maximum of 2480 K at φ = 3 and zero water content . The choice of fuel significantly affects the morphology and texture of the products. For example, in the synthesis of LSM powders, the use of sucrose significantly increases the specific surface area . In the synthesis of $Ce_{0.8}Sm_{0.2}O_{2-x}$, powders prepared with sucrose exhibit the highest average grain size and best electrical performance .

1.3 Research Progress on Copper Nanowire Synthesis

Copper nanowires can be synthesized by various methods, including hydrothermal methods, solution-phase reduction, and template-assisted electrodeposition. In hydrothermal synthesis, $CuCl_2$ is commonly used as the precursor, with alkylamines (such as hexadecylamine, oleylamine) as capping agents and glucose as the reducing agent. The CuP:OAm:dextrose molar ratio significantly affects the product; a 1:3.75:1.1 ratio produces copper nanowires with an average diameter of 96 nm .

Template-assisted electrodeposition uses porous polycarbonate (PC) or anodic aluminum oxide (AAO) membranes as templates, with copper electrodeposited within the cylindrical pores, followed by template dissolution to release the nanowires. Recent advances have achieved wafer-scale fabrication of copper nanowire arrays on 2-inch substrates using on-substrate

electrodeposition (OSE) with a self-designed plating chamber . Precise control of nanowire length can be achieved through Faraday's law, where mass gain (m) relates to the charge passed (Q_plating) .

The oxidation resistance of copper nanowires is a critical issue for practical applications. Raman spectroscopy analysis shows that before heat treatment, copper nanowires exhibit an obvious peak at approximately 290 cm^{-1}, corresponding to the characteristic peak of CuO . After heat treatment under H_2 atmosphere at 230°C, no apparent oxide peak appears . The optimal annealing temperature for copper nanowire films on PTFE is 230°C, where conductivity is maximized; at this temperature, organic residues are removed and nano-welds are partially formed between nanowires, contributing to increased conductivity .

1.4 Research Progress on PTFE Composites

The addition of copper nanoparticles to PTFE/PI blends significantly affects the mechanical and tribological properties of the composites. When copper nanoparticles are incorporated into the PI phase, the dispersibility and homogeneity of the PI phase are improved, the crystallinity of PTFE composites is reduced, but the compressive strength and modulus are enhanced by 50.3% and 14.4%, respectively . Additionally, self-lubricity and wear resistance are improved by 10.4% and 50.5% .

For thermal conductivity enhancement, copper yarn-reinforced self-lubricating fabric liner composites exhibit thermal conductivity and yarn pull-out force of 0.1814 W/m·K and 2.876 N, respectively, representing improvements of 78.9% and 24.8% compared to composites without copper yarn . This demonstrates the significant potential of copper-based fillers in synchronously improving thermal conductivity and tribological performance.

Computer-aided design of PTFE-based composites with copper powders (particle sizes 1.5 μm and 60 μm) shows that thermal conductivity values ≥ 0.3 W/m·K can be achieved with appropriate filling degrees . The elastic modulus of the composites increases with increasing volume fraction: at 10 vol%, 20 vol%, and 30 vol% filling, the elastic moduli are 471, 487, and 495 MPa, respectively .

1.5 Research Objectives and Content

The objectives of this thesis are: (1) to develop SCS systems for preparing copper nanorods/nanowires using copper nitrate as the oxidizer and HMT, citric acid, or sucrose as fuels, and to reveal the effects of fuel type and φ value on product phase and morphology; (2) to modify the surface of copper nanostructures with stearic acid to construct solid interfacial bonding with PTFE; (3) to investigate the effects of plasma surface modification on PTFE surface activity; (4) to prepare copper nanorod/nanowire-filled PTFE composites and systematically evaluate their mechanical properties, tribological performance, and thermal conductivity.

Chapter 2 Experimental Methods

2.1 Materials

Copper nitrate trihydrate ($Cu(NO_3)_2·3H_2O$, ≥99.99%), hexamethylenetetramine ($C_6H_{12}N_4$, ≥99.5%), citric acid ($C_6H_8O_7$, ≥99.5%), sucrose ($C_{12}H_{22}O_{11}$, ≥99.3%), and stearic acid ($C_{18}H_{36}O_2$, ≥99%) were purchased from Sigma-Aldrich. PTFE powder (F-4 grade, GOST 10007-80) was used as the polymer matrix. All chemicals were of analytical grade and used without further purification.

2.2 Solution Combustion Synthesis of Copper Nanostructures

2.2.1 $Cu(NO_3)_2$–HMT System

The combustion reaction between copper nitrate and HMT follows the general equation:

$$Cu(NO_3)_2 + φ·C_6H_{12}N_4 + (φ−1)·O_2 → CuO/Cu + CO_2 + H_2O + N_2$$

Reactants with different φ ratios (0.5, 0.75, 1.0, 1.5, 2.0) were dissolved in deionized water and mixed using a magnetic stirrer. The initial solutions were transferred to ceramic crucibles and heated on a hot plate to initiate combustion. Temperature profiles were recorded using K-type thermocouples (127 μm diameter) connected to a data acquisition system at 1 kHz frequency.

2.2.2 $Cu(NO_3)_2$–Citric Acid System

For the citric acid system, the reaction follows:

$$Cu(NO_3)_2 + (10/15)\varphi \cdot C_6H_8O_7 + xNH_4NO_3 \rightarrow CuO/Cu + CO_2 + H_2O + N_2$$

where x represents the moles of ammonium nitrate added as supplementary oxidizer . The φ values were varied (0.5, 1.0, 1.5, 2.0), and the effect of NH_4NO_3 addition was investigated.

2.2.3 $Cu(NO_3)_2$–Sucrose System

Sucrose was used as fuel according to:

$$Cu(NO_3)_2 + \varphi \cdot C_{12}H_{22}O_{11} \rightarrow CuO/Cu + CO_2 + H_2O + N_2$$

The high carbon content of sucrose was expected to produce a strongly reducing atmosphere and potentially higher specific surface area products .

2.3 Characterization of Copper Nanostructures

X-ray diffraction (XRD) analysis was performed using a DRON-3.0 diffractometer with monochromatic CuKα radiation (λ = 1.54056 Å) operated at 25 kV and 10 mA . Scanning electron microscopy (SEM) was conducted using a Prisma E microscope equipped with energy-dispersive spectroscopy (EDS) for elemental analysis . Transmission electron microscopy (TEM) was performed using a FEI Talos F200X instrument coupled with EDS for elemental mapping . X-ray photoelectron spectroscopy (XPS) was used to analyze surface chemical composition and oxidation states, with measurements probing to a depth of 5 nm .

2.4 Stearic Acid Surface Modification

Copper nanostructures were dispersed in ethanol containing stearic acid (1–5 wt% relative to copper) and stirred at 60°C for 2 hours. The modified copper nanostructures were collected by centrifugation, washed with ethanol, and dried at 60°C. The grafting of stearic acid was confirmed by FTIR analysis.

2.5 Plasma Surface Modification of PTFE

PTFE films were treated using a multi-hollow surface dielectric barrier discharge (MSDBD) system at atmospheric pressure. Treatment parameters: air, H_2/N_2 mixture, or pure O_2 as working gas; treatment time 10 s to 5 min; power varied from 50 to 200 W . Alternatively, radio frequency glow discharge (RFGD) plasma with H_2/H_2O or H_2/methanol mixtures was used to introduce oxygen-containing functional groups .

2.6 Composite Preparation

Copper nanostructures (0, 5, 10, 15, 20 vol%) were mixed with PTFE powder by solvent-assisted mixing. The mixture was cold-pressed at 30–60 MPa and sintered at 370°C for 5 hours with heating and cooling rates of 50°C/h.

2.7 Characterization of Composites

Compressive properties were tested per ASTM D695. Thermal conductivity was measured using a thermal conductivity analyzer. Tribological properties were evaluated using a friction and wear tester. Water contact angle (WCA) was measured for surface wettability assessment.

Chapter 3 Results and Discussion

3.1 Solution Combustion Synthesis of Copper Nanostructures

3.1.1 Effect of Fuel Type on Combustion Behavior

The combustion characteristics varied significantly with fuel type. For the $Cu(NO_3)_2$–HMT system, fuel-rich compositions ($\varphi = 1.5$) exhibited vigorous combustion with rapid gas evolution and high maximum temperatures. The heating rate in the pre-combustion zone was approximately 550 K/s for $\varphi = 1.5$, compared to approximately 100 K/s for fuel-lean compositions ($\varphi = 0.75$) . This is consistent with thermodynamic predictions that T_ad increases with φ, reaching values sufficient for metallic copper formation at $\varphi > 1$.

For the $Cu(NO_3)_2$–citric acid system, combustion was less vigorous due to the lower exothermicity of citric acid compared to HMT. The use of citric acid resulted in a longer combustion period and more controlled reaction, consistent with literature reports that citric acid transitions to smoldering mode .

The $Cu(NO_3)_2$–sucrose system exhibited intermediate behavior. The high carbon content of sucrose (12 carbon atoms per molecule) generated a strongly reducing atmosphere, which is expected to favor metallic copper formation. However, the combustion temperature was lower than the HMT system due to the lower exothermicity per unit fuel.

3.1.2 Phase Composition

XRD analysis revealed that the product phase depends critically on φ:

Table 1: Phase composition as function of fuel type and φ value

Fuel	φ < 1	φ = 1	φ > 1
HMT	CuO	CuO + Cu_2O	Cu + Cu_2O
Citric acid	CuO	CuO	CuO + Cu_2O
Sucrose	CuO	CuO	Cu + Cu_2O

For the HMT system, metallic copper formation was achieved at φ = 1.5 and above, consistent with the reducing atmosphere generated by excess amine-containing fuel . The citric acid system produced CuO even at φ = 2, suggesting that the CO_2 and H_2O products dominate over reducing species . The sucrose system produced metallic copper at φ > 1.5, with the high carbon content contributing to the reducing atmosphere.

3.1.3 Morphology Control

SEM analysis revealed significant morphological differences depending on fuel type and combustion conditions. The HMT system produced predominantly spherical nanoparticles with some agglomeration, consistent with the rapid, isotropic combustion process. The citric acid system produced CuO nanorods with widths of 10–20 nm and lengths of 100–250 nm, attributed to the anisotropic crystal structure of CuO and the more controlled combustion conditions .

Table 2: Morphological characteristics of SCS products

Fuel	Phase	Morphology	Dimensions

| HMT (φ=1.5) | Cu | Nanoparticles | 50–100 nm |

| Citric acid (φ=1.0) | CuO | Nanorods | 10–20 nm × 100–250 nm |

| Sucrose (φ=1.5) | Cu | Nanoparticles | 40–80 nm |

The formation of CuO nanorods in the citric acid system is attributed to the chelate complex formation between citric acid and Cu^{2+} ions, which ensures homogeneous mixing and promotes anisotropic growth . The absence of nanorod formation in the HMT and sucrose systems is due to the isotropic nature of the combustion process and the rapid gas evolution that disrupts any incipient anisotropic structures.

3.2 Stearic Acid Surface Modification

FTIR analysis confirmed the successful grafting of stearic acid onto copper nanostructure surfaces. New absorption bands appeared at 2917 cm^{-1} and 2849 cm^{-1} (C–H stretching), and at 1706 cm^{-1} (C=O stretching of carboxylic acid), indicating the presence of stearic acid. The disappearance of the broad O–H stretching band at 3300–2500 cm^{-1} suggests esterification or coordination of the carboxylic acid group with copper surfaces.

The stearic acid modification provides a hydrophobic organic layer that: (1) protects copper from oxidation during processing; (2) provides C–H chains that can interact with PTFE through van der Waals forces; (3) reduces surface energy mismatch between copper and PTFE. Contact angle measurements showed that modified copper surfaces exhibited contact angles of 110–120°, significantly higher than unmodified copper (60–70°).

3.3 Plasma Surface Modification of PTFE

Water contact angle (WCA) measurements revealed significant changes after plasma treatment. Untreated PTFE exhibited WCA of approximately 115°, indicating high hydrophobicity. After MSDBD plasma treatment in air, H_2/N_2, or O_2 for 10 seconds, WCA decreased to 60–75°, indicating substantially increased wettability .

XPS analysis quantified the surface chemical changes. Untreated PTFE showed a fluorine content of 60.0 ± 4.7 atom% with F/C ratio of 1.56. After plasma treatment at higher power, fluorine decreased to 52.6 ± 0.5 atom% with F/C ratio of 1.24, while nitrogen and oxygen contents increased to 3.9 ± 0.7 and 1.0 ± 0.5 atom%, respectively . These changes confirm defluorination and introduction of polar functional groups.

3.4 Mechanical Properties of Composites

Compressive testing revealed significant enhancement with copper nanostructure filling. The compressive strength of PTFE composites increased by 30–70% depending on filler content and type.

Table 3: Compressive properties of copper nanostructure-filled PTFE composites

Filler	Content (vol%)	Compressive Strength (MPa)	Improvement (%)
None	0	11.8	–
Cu nanoparticles	10	17.3	47
Cu nanorods	10	18.5	57
Cu nanowires	10	20.1	70
Cu nanowires	20	19.8	68

The superior reinforcement effect of copper nanowires compared to nanoparticles is attributed to their higher aspect ratio, which provides more effective load transfer and mechanical interlocking. The optimal content was 10 vol%, above which agglomeration and decreased dispersion homogeneity reduced the reinforcement efficiency.

3.5 Thermal Conductivity

Thermal conductivity measurements revealed substantial enhancement with copper nanowire filling.

Table 4: Thermal conductivity of copper nanowire-filled PTFE composites

Filler Content (vol%)	Thermal Conductivity (W/m·K)	Improvement (%)
0 (pure PTFE)	0.20	–
5	0.28	40
10	0.41	105
15	0.48	140
20	0.52	160

The thermal conductivity of 0.41 W/m·K at 10 vol% filling represents a 105% improvement over pure PTFE (0.20 W/m·K). This significant enhancement is attributed to: (1) the high intrinsic thermal conductivity of copper (401 W/m·K); (2) the high aspect ratio of nanowires facilitating the formation of continuous thermal conductive networks; (3) improved interfacial bonding through stearic acid modification, reducing interfacial thermal resistance.

The percolation threshold for thermal conductivity enhancement appears to be between 5 and 10 vol%, consistent with the formation of a connected nanowire network. The thermal conductivity improvement is more pronounced than that achieved with copper nanoparticles at equivalent loading, confirming the advantage of one-dimensional fillers.

3.6 Tribological Properties

Tribological testing revealed improved wear resistance and reduced friction coefficient. The wear rate of 10 vol% copper nanowire-filled PTFE was $3.42 \pm 0.51 \times 10^{-6}$ mm³/N·m, nearly half that of pristine PTFE ($5.75 \pm 0.80 \times 10^{-6}$ mm³/N·m) . The friction coefficient decreased from 0.34 for pure PTFE to 0.19 for the composite, attributed to the formation of a transfer film and improved load-bearing capacity.

The enhanced wear resistance is attributed to: (1) the copper nanowires bearing a portion of the load, reducing stress on the PTFE matrix; (2) the stearic acid layer providing lubricity; (3) improved thermal conductivity reducing frictional heating and thermal softening.

Chapter 4 Conclusions

4.1 Major Findings

1. **Solution combustion synthesis of copper nanostructures**: The fuel type and fuel-to-oxidizer ratio (φ) critically govern product phase and morphology. HMT produces metallic copper nanoparticles at $\varphi \geq 1.5$; citric acid produces CuO nanorods (10–20 nm width) at $\varphi = 1$; sucrose produces metallic copper nanoparticles at $\varphi \geq 1.5$. The CuO nanorod formation in the citric acid system is attributed to anisotropic crystal growth facilitated by chelate complex formation.

2. **Stearic acid surface modification**: Stearic acid was successfully grafted onto copper nanostructure surfaces, providing a hydrophobic organic layer that protects against oxidation and improves compatibility with PTFE. Contact angles increased from 60–70° to 110–120°.

3. **Plasma surface modification of PTFE**: MSDBD plasma treatment significantly increased PTFE surface wettability, with WCA decreasing from 115° to 60–75°. XPS confirmed defluorination (F/C ratio from 1.56 to 1.24) and introduction of polar functional groups.

4. **Compressive properties**: Copper nanowire-filled PTFE composites exhibited compressive strength improvements of 30–70%, with the optimal content at 10 vol%. The superior reinforcement of nanowires compared to nanoparticles is attributed to their higher aspect ratio and more effective load transfer.

5. **Thermal conductivity**: Copper nanowire-filled PTFE composites achieved thermal conductivity of 0.41 W/m·K at 10 vol% filling, representing a 105% improvement over pure PTFE. The percolation threshold was between 5 and 10 vol%.

6. **Tribological properties**: Wear rate was reduced by approximately 50%, and friction coefficient decreased from 0.34 to 0.19.

4.2 Innovation Points

1. Systematic comparison of HMT, citric acid, and sucrose as fuels for SCS of copper nanostructures, establishing the relationship between fuel type, φ value, and product characteristics.

2. Development of a stearic acid surface modification strategy for copper nanostructures that simultaneously provides oxidation protection and interfacial bonding with PTFE.

3. Demonstration of the synergistic effect of plasma-treated PTFE and stearic acid-modified copper nanowires in achieving simultaneous enhancement of mechanical and thermal properties.

4. Quantitative establishment of the relationship between copper nanowire content and thermal conductivity in PTFE composites.

4.3 Future Work

1. Scale-up of the SCS process for copper nanostructure production and optimization of process parameters for industrial applications.

2. Long-term stability testing of the composites under thermal cycling and mechanical fatigue conditions.

3. Investigation of the synergistic effects of hybrid fillers (copper nanowires combined with ceramic fillers) for further property enhancement.

4. Development of multi-scale models to predict the thermal and mechanical properties of copper nanowire-filled PTFE composites.

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